

Prompt-Based Interpretability in Dialogue Models: Evaluating Faithfulness and Reasoning under Low-Resource Conditions

Our Problem

Core Challenge

Large Language Models (LLMs) in safety-critical domains

like medical/finance must be accurate and their reasoning must be faithful, and the explanation should truly reflect the model's decision process.

Our Question

How does **structured prompting** impact both task performance and interpretability in specialized, low-resource dialogue?

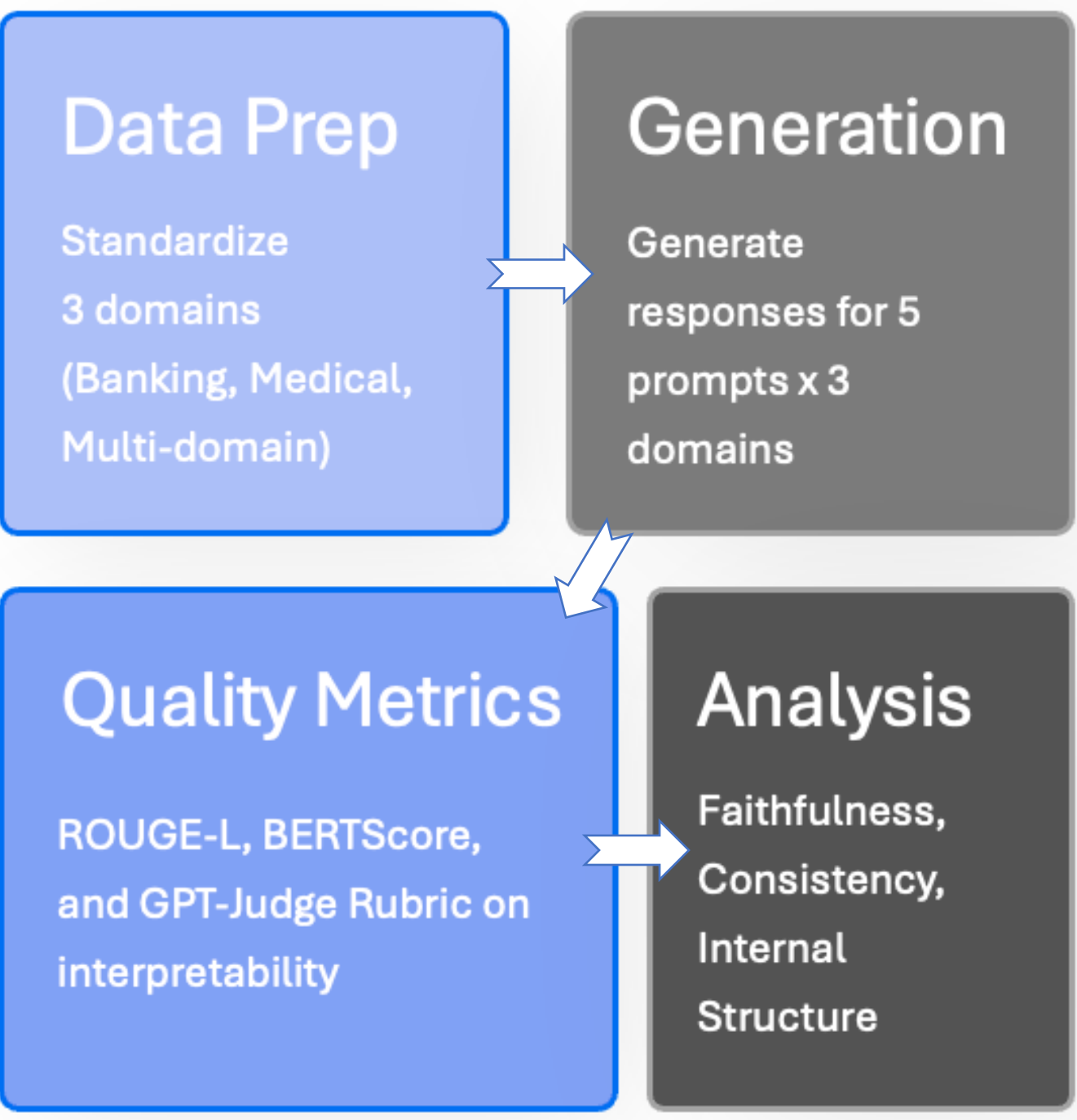
Methods: Prompt Strategies & Pipeline

Five Prompt Families

We evaluate **Mistral-7B-Instruct-v0.2** generator under 5 prompt families in 3 distinct dialogue domains (Banking, Medical, Multi-domain):

Prompt Family	Rationale
Baseline	Minimal instructions
Chain-of-Thought (CoT)	Step-by-step thinking
Self-Explain	Requires brief justification of final answer
Domain-Aware	Incorporates explicit domain guidelines
Critic	Requires self-critique/review before outputting the final answer

Experiment Pipeline



Key Findings: Task Quality Evaluation

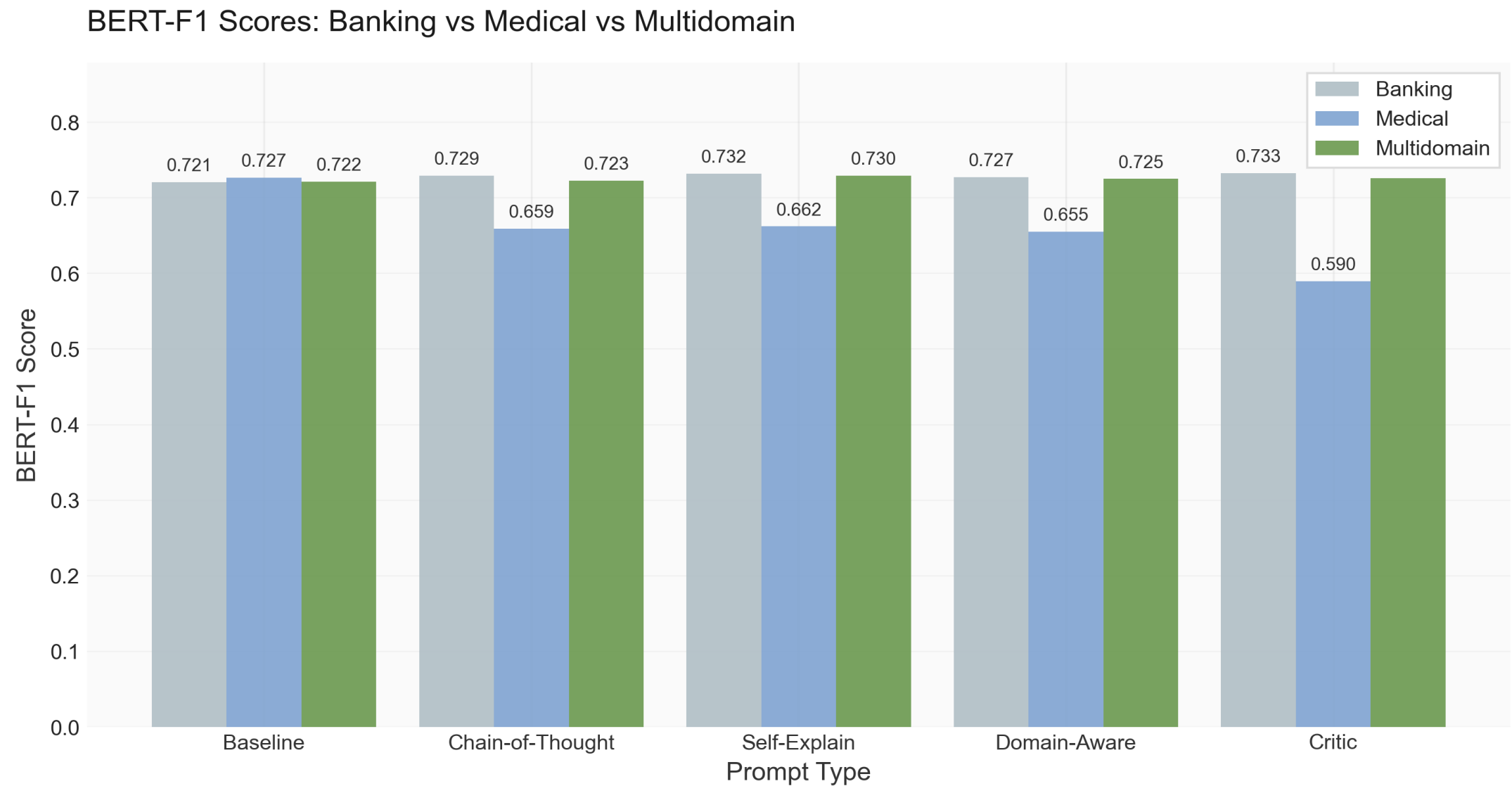
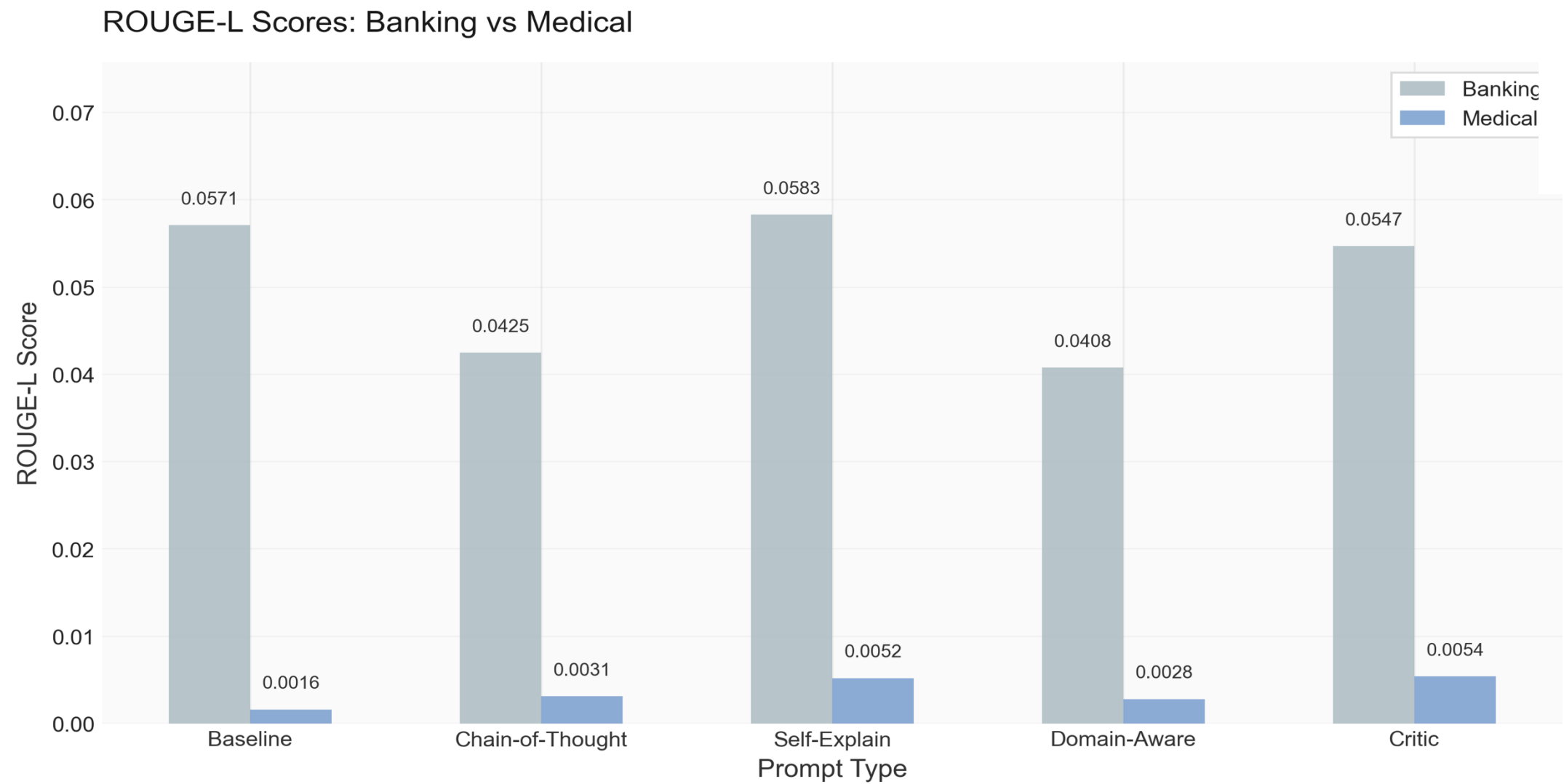
Overall Task Quality Metrics on All 3 Domains

Prompt Family	BERT-F1	ROUGE-L
Self-Explain	0.7233	0.0226
Critic	0.6762	0.0231
Chain-of-Thought (CoT)	0.7008	0.0175
Domain-Aware	0.6991	0.0168
Baseline	0.7047	0.0244

ROUGE-L scores vary by domain: Banking shows moderate overlap; Medical and Multidomain are near zero, suggesting different phrasing from references. **BERT-F1 scores are consistently high across domains**, indicating strong semantic similarity despite low lexical overlap.

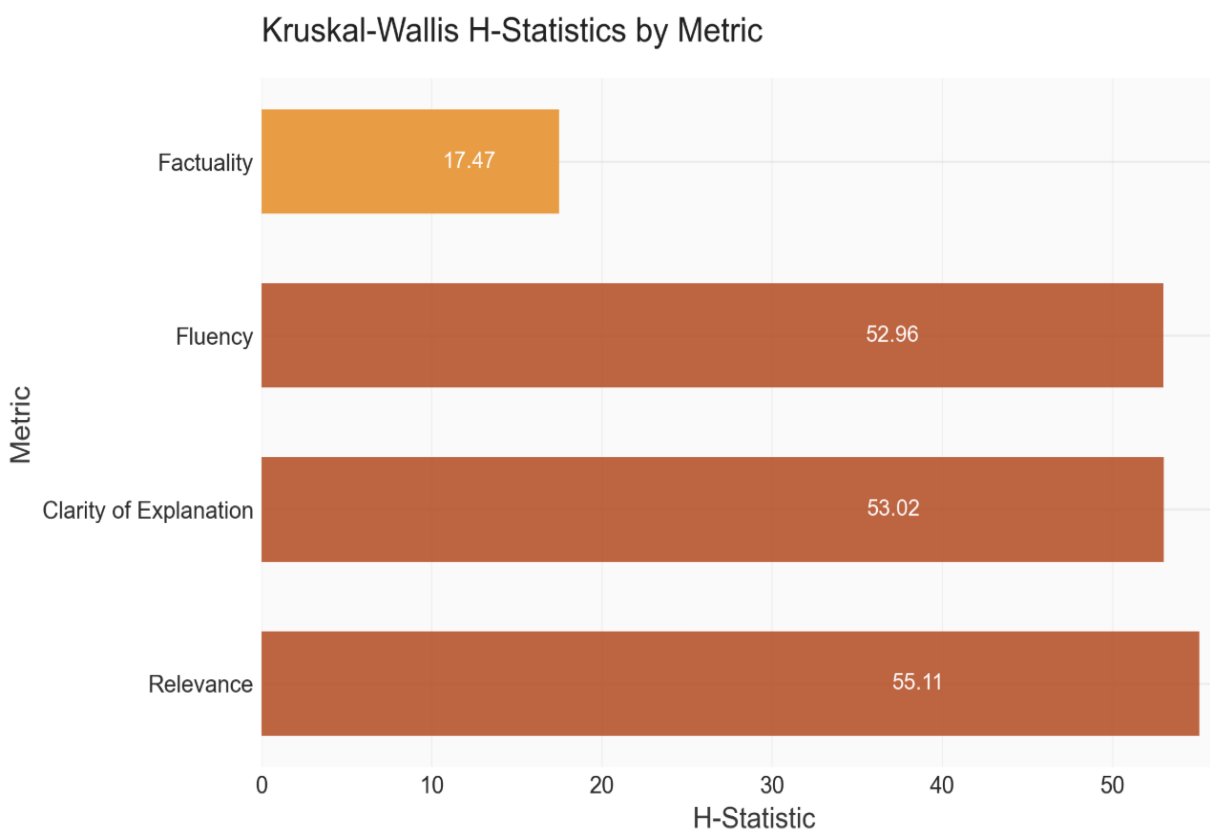
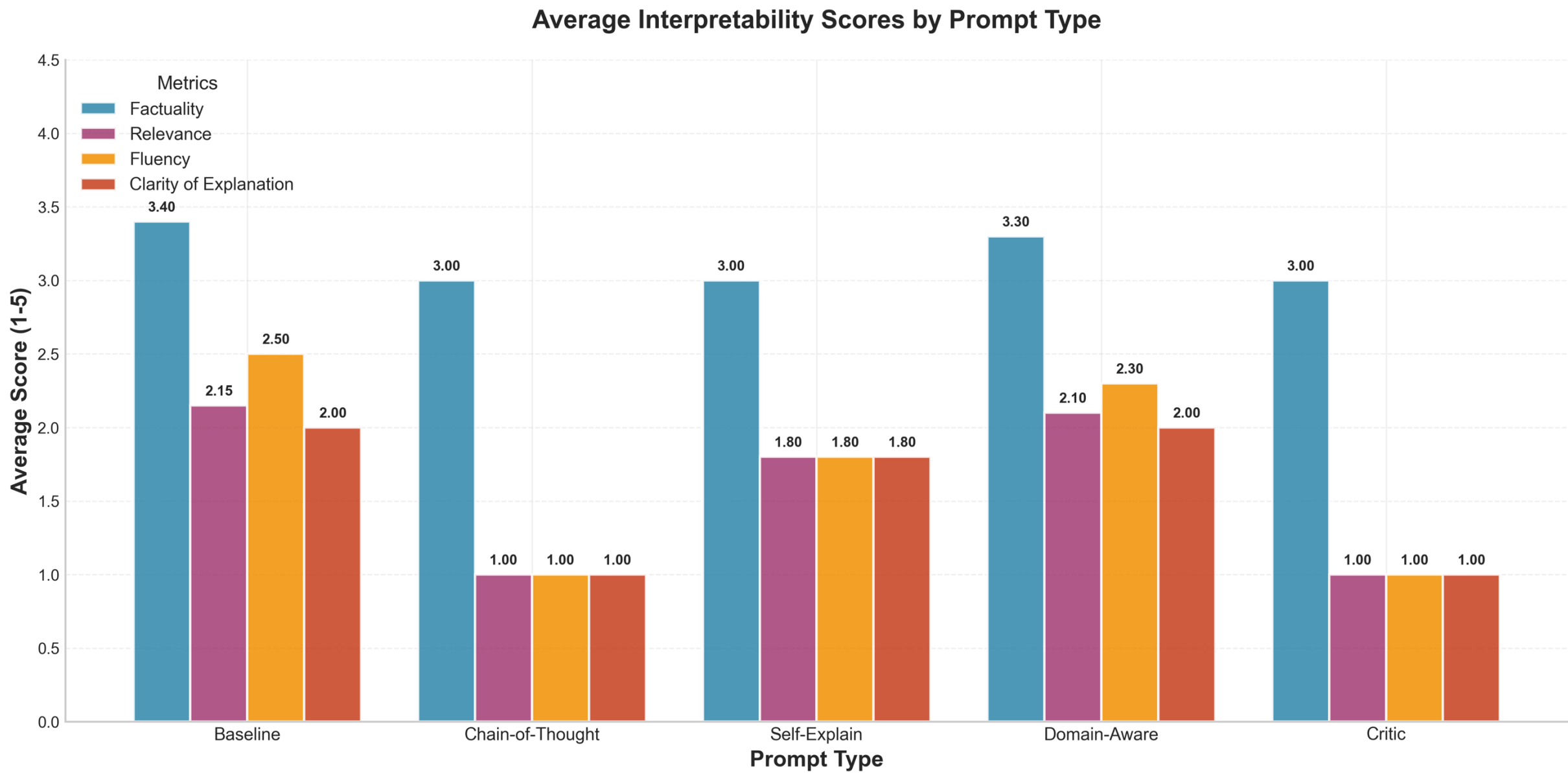
Prompt type patterns

- Self-explain & Critic:** perform better on **lexical overlap** (ROUGE-L)
- Baseline & Self-Explain:** stronger **semantic similarity** (BERT-F1)
- Domain-aware & Chain-of-Thought:** more **variability**



Deeper Analysis: Interpretability Evaluation

Interpretability Evaluation Using gpt-5.1 model on Banking Domain



- Baseline & Domain-Aware:** best interpretability
- CoT & Critic:** lowest scores across metrics
- Factuality:** high and stable for all prompts
- Largest gaps in **Relevance, Fluency, Clarity**
- Kruskal-Wallis:** strong prompt-type effect ($p < 0.001^*$)

→ **Takeaway:** Simple or domain-guided prompts **outperform** structured reasoning

Conclusion & Future Directions

- Structured rationales (Self-Explain)** give the best **semantic quality** (highest BERT-F1).
- Domain-Aware & Baseline** give the best **interpretability** (Relevance, Fluency, Clarity).
- CoT & Critic** consistently underperform in interpretability for low-resource dialogue.

Key Takeaway

- Prompt structure has a **strong, measurable impact** on both quality and interpretability.
- More reasoning does **not** guarantee clearer or safer response
- For safety-critical dialogue, **simple or domain-guided prompts** are more reliable than heavy reasoning prompts.
- Interpretability can **degrade** when prompts force models into doing **unnecessary reasoning** steps.

Future Work

- Explore **fine-tuning** with self-explanations to integrate interpretability directly into model.
- Test structured prompting on **higher-resource** or **multi-turn** dialogues.

References

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